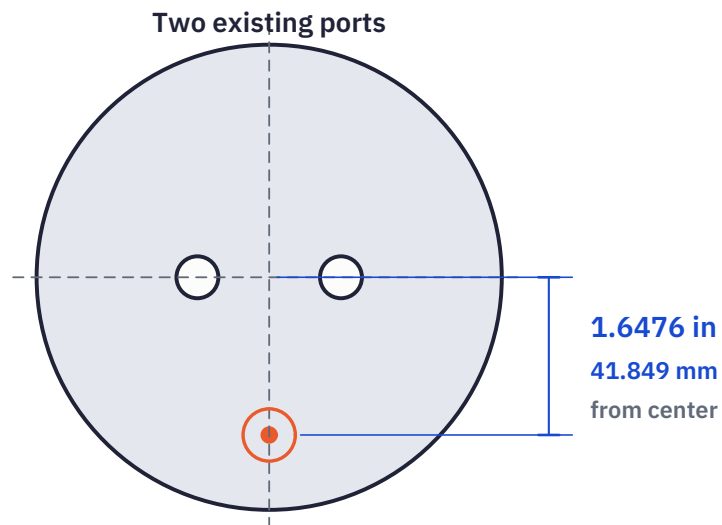


DRILL THE BLIND REGISTER

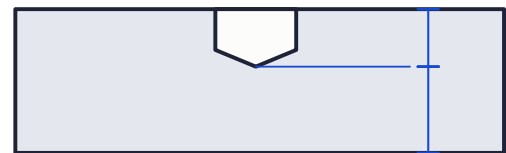
One small pocket, inside each endcap.
It locates the float rod without piercing the plate.

INSIDE FACE / TOP VIEW



ENLARGED SECTION

Inside face



9/64 in drill
0.100 in to TIP
0.150 in stays solid
Do not drill through.

1 Locate the pocket



Inside face up. Mark the location shown above on both plates.

2 Clamp the disc



Secure it to the press table. Check the bit and chuck clear every clamp.

3 Set + prove the stop



Touch the stopped tip to the face. Set 0.100 in travel; prove on scrap.

4 Drill both plates



9/64 in M35 cobalt, 135° split point. ~740 RPM;
Tap Magic on the point.

CHECK BOTH POCKETS

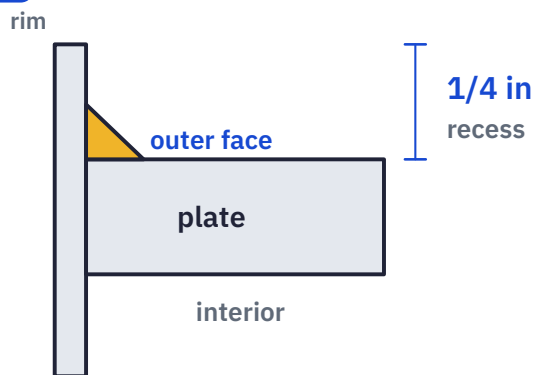
Blind, 0.100 in to tip; 0.150 in of solid plate remains.

SET UP THE CLOSURE WELD

Working end up. Endcap's outer face toward you.
The register face and float rod face into the tube.

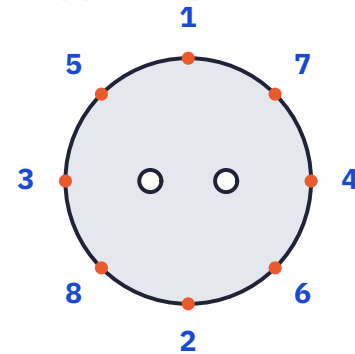
Class 4 laser. Laser-rated eye/face protection, gloves, protective clothing.
Use a protected welding area with fume extraction.

1 Clean + seat



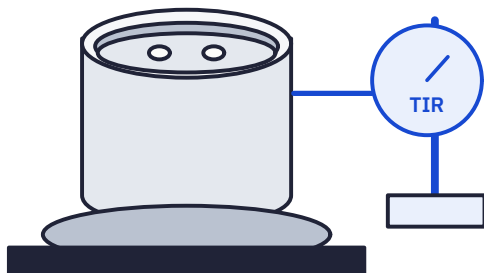
Clean the bore band and plate's outer face.
Deburr tube ID + OD.
Seat on the 1/4 in spacer; check all around.

2 Tack opposite pairs



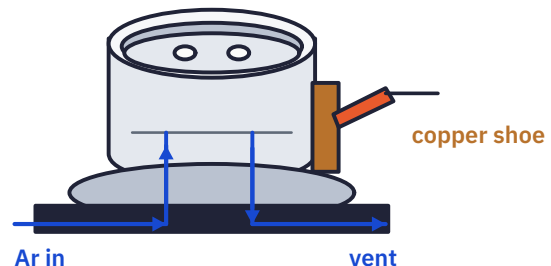
Eight tacks, bisecting the gaps in opposite pairs.
Recheck seated depth.
The rod must not hold the plate up.

3 Indicate the work



Clamp the base; touch adjusters lightly.
Weld-end radial TIR ≤ 0.25 mm.
Endcap-face TIR ≤ 0.30 mm.
Use the commissioned rotator.

4 Purge + prove contact



First: purge through open held end.
Second: lower port in; other port vent.
Scuff tube + shoe; clip work lead to shoe.
Laser OFF: continuity for 2 dry turns.

POWER

60%

WOBBLE

80 Hz / 2 mm

WIRE

12 mm/s

ARGON

2 s pre/post

FILLER

ER316L .030

Recorded practice recipe. Closure root fusion and internal oxidation still need qualification. These settings are for the endcap closure.

START THE BEAD

Your foot turns the tube. Your trigger starts the weld.
Hold the head steady as the joint travels under it.

1 Set speed while stopped

status
speed 8.0



Start at 8 mm/s travel. Verify direction from above and return the index to the first tack.

2 Aim into the corner

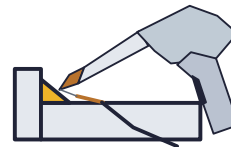


plate mass

Beam into plate mass, washing onto wall. Wire on arriving side. Use the coupon-proven angle + standoff.

3 Ready the welder

WELDING

Wire feed

Gas

Laser enable

PPE + protected area ready; shielding and back-purge established. Enable wire feed and laser on the X1 Pro.

4 Press + hold the pedal

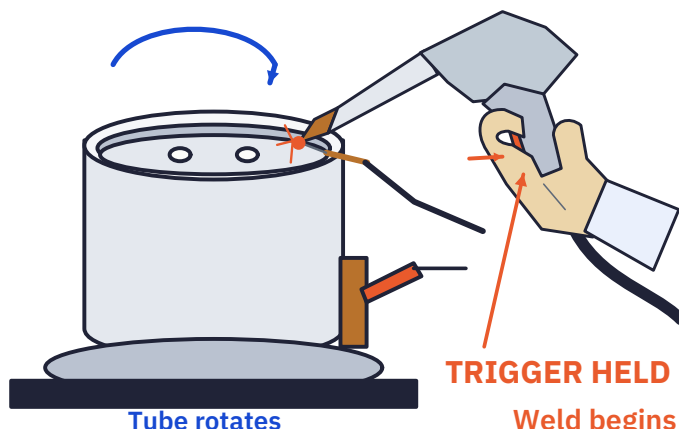


Keep the trigger released. Hold the pedal down until rotation is steady; keep the head at the joint.

KNOW THE FINISH BEFORE YOU START

Carry the bead a full lap, then about 20° past the first tack. Trail off; **release trigger, then pedal**. Keep the head in position for post-flow. The table has no automatic stop. Snip stuck wire with the gun held still.

5 Pedal held. Pull + hold the trigger.



STEADY ROTATION FIRST

